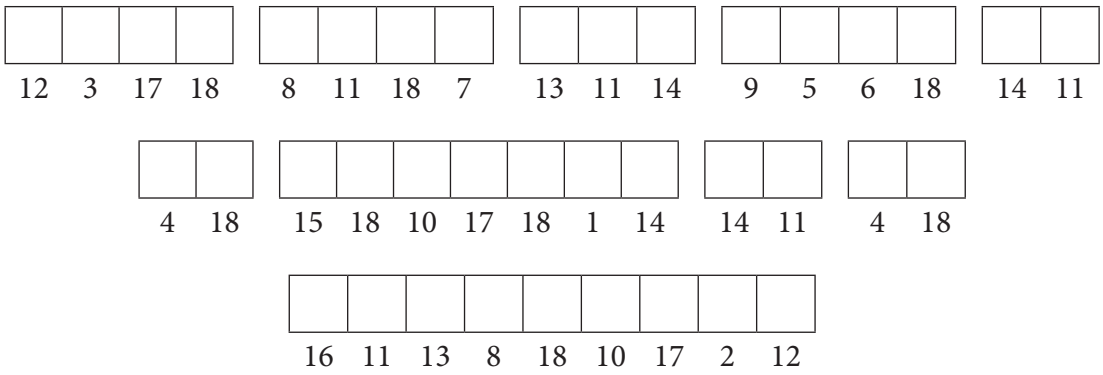
Name: _____

WORKSHEET

Geometric progressions 1



Annette Funicello

Rule

In a geometric progression, the nth term t_n can be found by $t_n = ar^{n-1}$, where a is the first term and r is the common ratio.

The common ratio $r = \dfrac{t_n}{t_{n-1}}$

Questions

What is the common ratio in the following geometric progressions?

1 4, 12, 36, 108, . . .

2 24, −12, 6, −3, . . .

3 5, $10\sqrt{3}$, 60, $120\sqrt{3}$, …

What is the rule for t_n for the following geometric progressions?

4 3, 15, 75, 375, . . .

5 192, −38.4, 7.68, −1.536, . . .

6 0.05, 0.3, 1.8, 10.8, . . .

7 What is the 12th term of $\dfrac{3}{25}, -\dfrac{3}{5}, 3, -15, \ldots$?

8 What is the 7th term of $5i, -5, -5i, 5, \ldots$?

9 What is the 50th term of 80 000, 76 000, 72 200 . . .

Solutions

A $192 \times (-0.2)^{n-1}$

B $3 \times 5^{n-1}$

C 3

D $-5i$

E $320x^6$

F −262 144

H 6479.58

I $2\sqrt{3}$

L 7

N 14

O 6

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10 A person begins to train for 10 km race by running 1 km on the first day and increasing this distance by 10% each day. After how many days will be run at least 10 km?

11 A car depreciates by 20% each year. After how many years will a car bought for \$28000 be worth less than \$10 000?

How many terms are in the following geometric progressions?

12 500, 50, 5, . . . , 0.0005

13 $-\frac{3}{4}, 1\frac{1}{2}, -3, \ldots, 6144$

14 $5\sqrt{2}, 20, 40\sqrt{2}, \ldots, 335\,544\,320$

15 20, 21, 22.05, . . . , 2504.7859

16 Find the 10th term of a geometric progression if $t_3 = 180$ and $t_7 = 2\frac{2}{9}$.

17 Find t_{20} of a geometric progression if $t_2 = -1, t_5 = 8$.

18 Find the t_4 of a geometric progression if $t_4 = 40x^3, t_{11} = 5120x^{10}$.

P 100

R 26

S $-5\,859\,375$

T 18

U $-\frac{1}{2}$

V $0.05 \times 6^{n-1}$

W $\frac{20}{243}$

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